

CS201 Completion

10 June 2024

Olcay Taner YILDIZ

I. QUESTION (ALGORITHM ANALYSIS)

```

1  sum = 0;
2  for (i = 1; i <= N; i++)
3      for (j = i; j <= N - i; j++)
4          sum++;

```

What is sum in terms of N ? Show your work.

II. QUESTION (LINKED LIST)

Write the method

```
void deleteEven()
```

which removes the nodes with even values in the original list. Your method should run in $\mathcal{O}(N)$ time.

III. QUESTION (TREE)

Write a recursive method

```
int inBetween(int k1, int k2)
```

that returns the number of all nodes in the binary search tree whose key is between k_1 and k_2 . Your method should run in $\mathcal{O}(\log N + K)$ time, where N is total number of nodes and K is the number of nodes which have values between k_1 and k_2 in the tree.

IV. QUESTION (GRAPH)

A graph represents a ring topology if all the nodes create a circular path. Each node is connected to two others, like points on a circle. Write a method for linked list implementation

```
bool isRing()
```

which returns true, if the graph is ring topology, false otherwise. Your method should run in $\mathcal{O}(E)$ time.

V. QUESTION (SORTING)

Given an array of N numbers write a function that determines the maximum length of the sorted sub list in the array. A sublist start from index i and continues with indexes $i+1$, $i+2$, ..., $i+k$. Your function should run in $\mathcal{O}(N)$ time.

```
int maxSortLength(int* A, int N)
```